



Technology Data Sheet (TDS)

TINMORRY PET-CF

Version 1.0

• Basic Introduction

TINMORRY PET-CF is a carbon fiber reinforced PET composite. Its formula has been specially designed, tested and optimized to provide excellent processing stability, printing stability and comprehensive mechanical properties. Compared with generic PETG-CF and ABS-CF, it has higher strength and stiffness. And compared with PA6-CF, it is less likely to warp, and does not get soft after absorbing water and thus can be used in humid air or water for a long time. Besides, the carbon fibers inside bring special texture and finish, which minimize the layer lines and make the prints surface more delicate and consistent. Moreover, being compatible with most FDM printers on the market (hardened steel nozzles are recommended), available for various colors and cost-efficient, TINMORRY PET-CF is really an ideal entry-level fiber-reinforced engineering filament.

• Specifications

Subjects	Data
Diameter	1.75 ± 0.03 mm
Net filament weight	1000 g
Length	315 ± 5 m
Spool material	PC ((Temperature resistance: ~ 120 °C)
Spool size	External Diameter: 198 mm; Inner Diameter: 56 mm; Width: 65 mm;

• Recommended Printing Settings

Subjects	Data
Drying settings before printing	80 °C, ≥ 8 h (Blast drying oven, or filament drying box)
Printing temperature and humidity	≤ 90 °C, ≤ 20% RH (Put in a working filament drying box, or put in an air-tight container and sealed with desiccant)
Storage temperature and humidity	≤ 30 °C, ≤ 20% RH (Sealed with desiccant)
Compatible support material	Itself or support filament for PET
Compatible printer type	Open-frame, enclosed-frame
Compatible nozzle material	Hardened Steel
Compatible nozzle size	0.4, 0.6 (recommended), 0.8 mm
Compatible plate type	Textured PEI Plate or other plate
Plate surface preparation	Glue
Nozzle temperature	260 - 290 °C
Bed temperature	90 - 100 °C (depending on the printer and plate types)
Printing speed	≤ 150 mm/s
Chamber temperature	≤ 60 °C

• Properties

The commonly used physical properties, mechanical properties, chemical properties, printing properties and other properties of TINMORRY PET-CF have been tested and are shown in the tables below.

• Physical Properties

Subjects	Testing Methods	Data
Density	ISO 1183	1.32 g/cm ³
Saturated water absorption rate	25 °C, 55% RH	0.34%
Melt index	270 °C, 2.16 kg	22.5 ± 2.2 g/10 min
Glass transition temperature	DSC, 10 °C/min	78 °C
Crystallization temperature	DSC, 10 °C/min	N / A
Melting temperature	DSC, 10 °C/min	252 °C
Vicat softening temperature	ISO 306, GB/T 1633	91 °C
Heat deflection temperature 1	ISO 75, 1.82 MPa	84 °C
Heat deflection temperature 2	ISO 75, 0.45 MPa	89 °C

• Mechanical Properties

Properties	Testing Methods	Data
Tensile strength (XY)	ISO 527, GB/T 1040	78 ± 6 MPa
Tensile strength (Z)	ISO 527, GB/T 1040	36 ± 3 MPa

Young's modulus (XY)	ISO 527, GB/T 1040	2382 ± 196 MPa
Young's modulus (Z)	ISO 527, GB/T 1040	1753 ± 139 MPa
Breaking elongation rate (XY)	ISO 527, GB/T 1040	7.6 % ± 1.6%
Breaking elongation rate (Z)	ISO 527, GB/T 1040	4.8 % ± 1.7%
Bending strength (XY)	ISO 178, GB/T 9341	108 ± 6 MPa
Bending strength (Z)	ISO 178, GB/T 9341	49 ± 4 MPa
Bending modulus (XY)	ISO 178, GB/T 9341	3081± 171 MPa
Bending modulus (Z)	ISO 178, GB/T 9341	2137 ± 185 MPa
Impact strength (XY)	ISO 179, GB/T 1043	38.5 ± 2.4 kJ/m ²
Impact strength (Z)	ISO 179, GB/T 1043	5.9 ± 1.2 kJ/m ²

• Chemical and Other Properties

Subjects	Data
Component	PET, carbon fiber
Skin irritation	May cause allergy in some people, and wearing gloves is suggested
Resistance to water	Yes
Resistance to oil and grease	Resistant to most kinds of daily oil and grease
Resistance to organic solvent	Not resistant to some organic solvents
Resistance to weak acid	Yes
Resistance to strong acid	Not resistant
Resistance to weak alkali	Yes
Resistance to strong alkali	Not resistant
Flammability	Yes
Odor during printing	Odorless
Odor during combustion	Pungent

• Specimen Info

Subjects	Data
Size	Shown at the following pictures
Nozzle temperature	275 °C
Bed temperature	90°C
Printing speed	150 mm/s
Infill density	100%
Infill pattern	Concentric

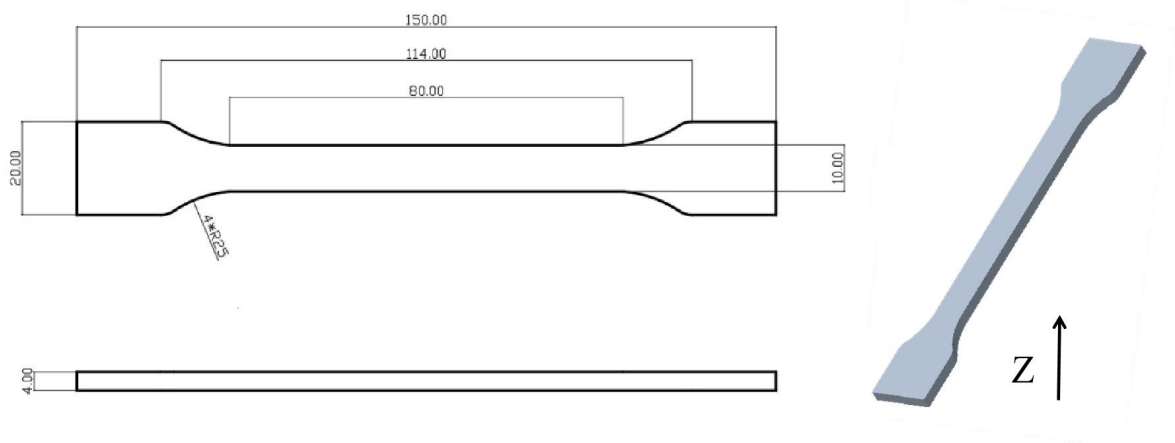
Statements:

1. The test specimens were printed under the key parameters mentioned above. Before testing, all specimens were annealed at 80 °C for 8 hours and then placed in an indoor environment at about 25 °C for 12 hours. Please note that different ambient temperatures, fan speeds and layer times can lead to varying cooling, which can affect the mechanical properties of the specimens, especially those in the Z-direction. Additionally, different printers can also result in variations in printing quality.
2. If high printing quality is required, it is recommended to dry every kind of TINMORRY

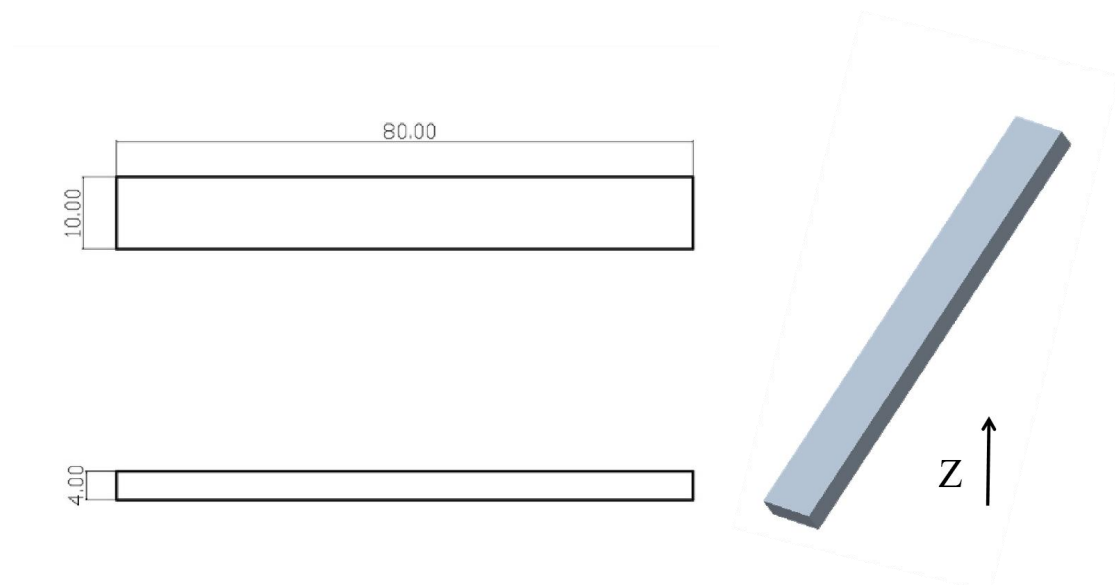
filaments before printing and use an air-tight container and desiccant to protect them from moisture during the printing process (with relative humidity inside the container less than 20%) to reduce the risk of issues related to moisture absorption in filaments. The recommended drying conditions are shown above. Note that when drying them, avoid placing the spools near the heater to prevent damage to the spools or the filaments due to excessive temperature. Do not use a kitchen oven or microwave oven.

3. TINMORRY PET-CF is a kind of crystalline material, and the strength, stiffness and heat resistance of its prints can be increased, but the toughness may be decreased. And some prints may undergo shrinkage or deformation after the annealing process. Note that the annealing effect is also affected by the size, structure, wall thickness, and infill density of the prints themselves. If you want to anneal them, it is recommended to use a blast drying oven which has uniform temperature, and it is recommended to first anneal at about 70°C for about 4 hours, then slowly increase the temperature to about 80°C and keep it for about 3 to 6 hours; if the prints are deformed halfway, it should be stopped immediately and a lower temperature should be used instead.

1. Tensile Test



2. Bending Test and Impact Test



• **Disclaimer**

The above properties data is obtained by TINMORRY through testing with standard samples, standard methods, and specific equipment, and is for reference and comparison only; if some data are changed, TINMORRY will update this TDS, but may not be able to inform the users, and please forgive us. The actual performance of 3D prints depends not only on the performance of the filaments used, but also on many factors, such as the degree of moisture absorption of the filaments, the printer, environmental conditions, model characteristics, printing parameters, etc.. When using TINMORRY FDM 3D printing filaments, users are responsible for the legality, safety, and performance of the prints. TINMORRY is responsible for the quality of our filaments, but not for any damage or injury that occurs during using them, or the usage scenarios of them.